

Predicting Customer Churn and Credit Default

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GitHub Repository:

https://github.com/pywen1004/DSC1107-SEC-1_SA2-GROUP-14_SANTIAGO

1. Introduction

This project aims to evaluate and compare various machine learning algorithms to predict customer churn and credit card defaults. By leveraging the Orange Telecom Churn and ISLR Default datasets, the primary objective is to identify the most effective models for accurately classifying customers who will leave a service and individuals who will default on their credit. The analysis compares the performance of regression-based models, tree-based models, and neural networks to determine the optimal balance between predictive power, handling of non-linear relationships, and real-world interpretability.

2. Methodology

2.1 Data Familiarization and Preparation

The datasets initially consisted of 3,333 rows and 20 columns for the Churn dataset, and 10,000 rows and 4 columns for the Default dataset. To prepare the data for modeling, several preprocessing steps were executed. First, non-predictive identifiers, specifically the 'State' and 'Area code' columns for the telecom data, were removed to prevent the models from learning irrelevant patterns. The target variables, 'Churn' and 'default', along with categorical features such as 'International plan', 'Voice mail plan', and 'student', were converted into binary numeric formats (0 and 1). Finally, to ensure algorithms like Logistic Regression and Neural Networks

functioned optimally, the numerical features were standardized using a standard scaler so that all variables contributed equally to the model's distance calculations.

2.2 Exploratory Data Analysis (EDA)

An initial exploration of the data revealed a significant class imbalance in both datasets. Specifically, the majority "No Churn" class represented approximately 85.5% of the telecom dataset, while the "No Default" class represented 96.67% of the credit dataset. The minority classes of interest represented only 14.5% and 3.33%, respectively. This imbalance highlighted the need to evaluate models using metrics beyond simple accuracy, such as Precision, Recall, and the F1-Score. Additionally, a correlation analysis was performed to identify multicollinearity, such as the perfect correlation between total day minutes and total day charges.

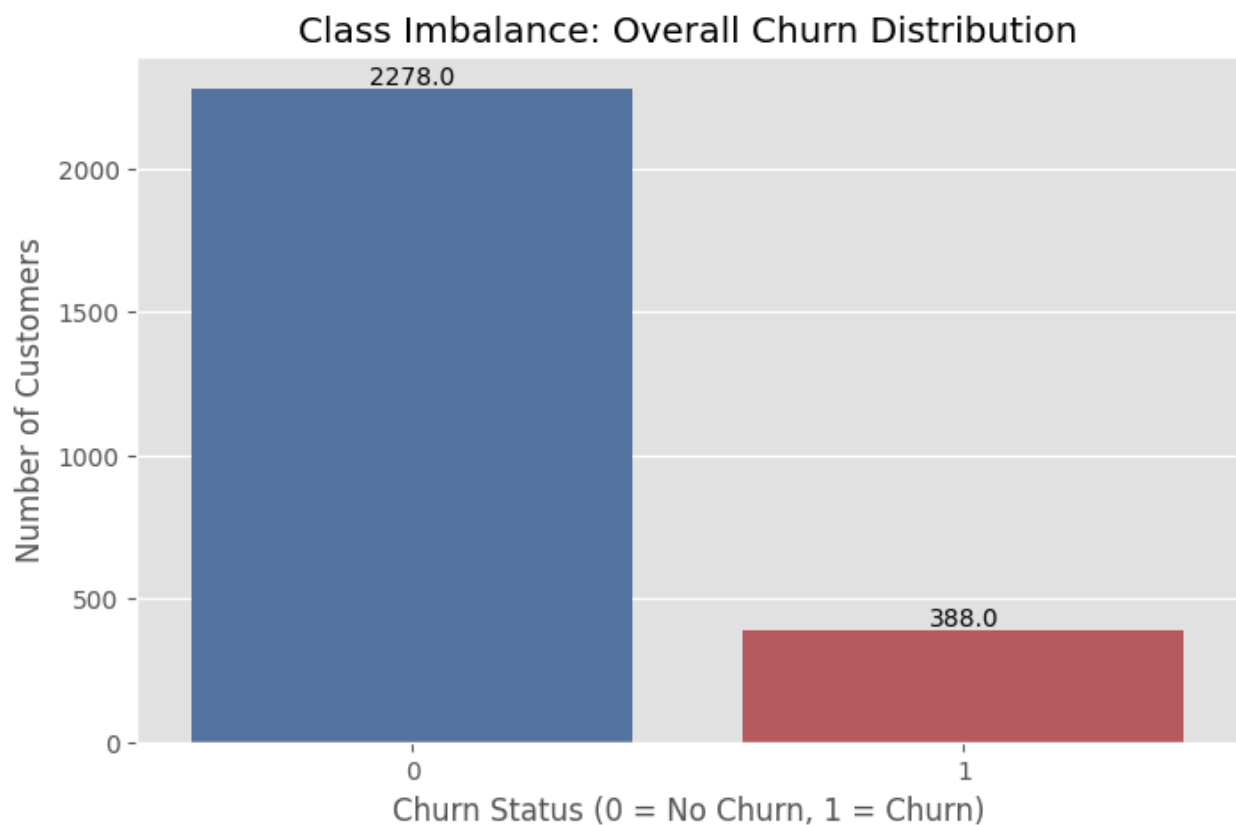


Figure 1: Distribution of the target variables highlighting the class imbalance.

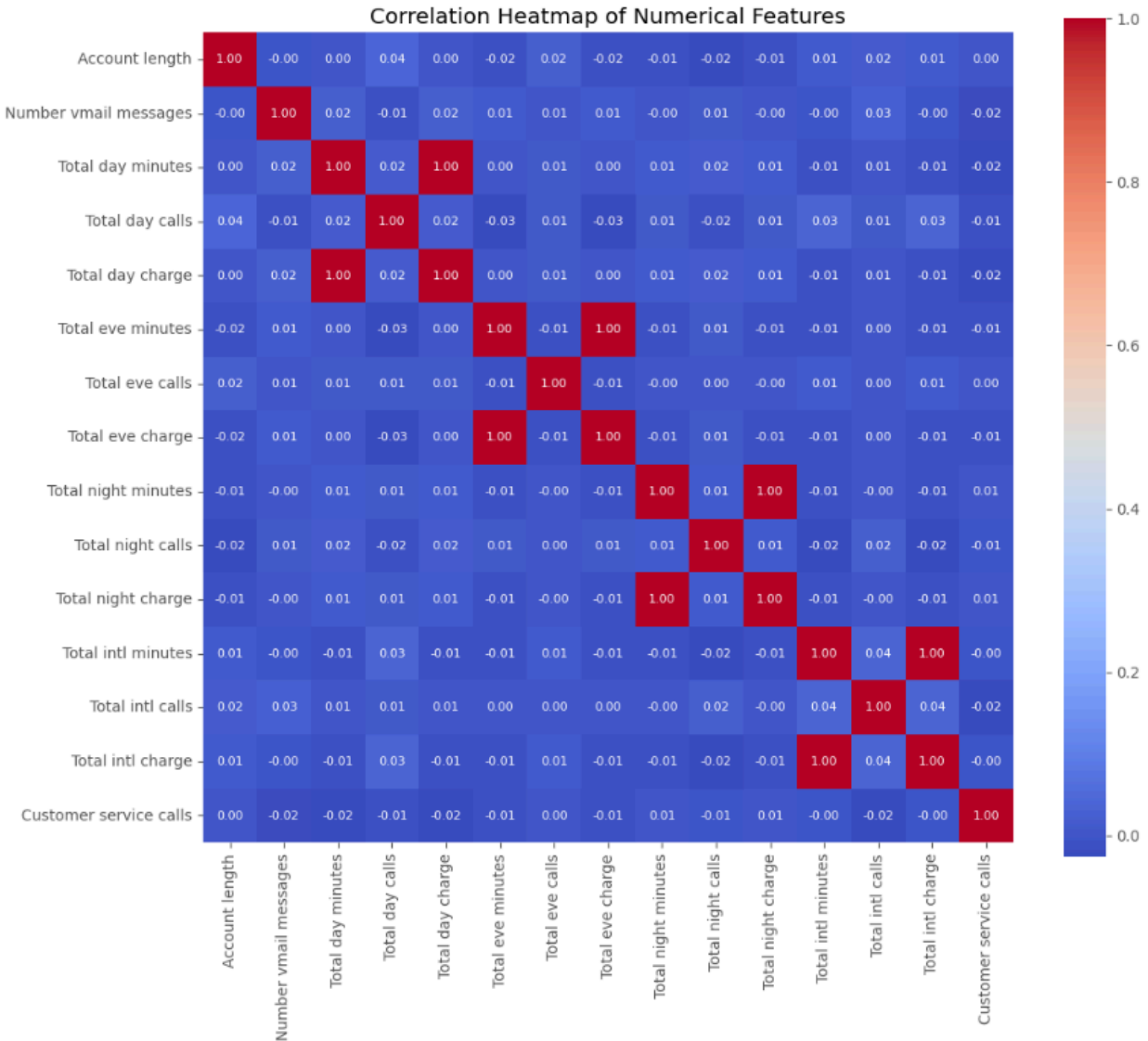


Figure 2: Correlation matrix highlighting relationships between numerical features.

2.3 Modeling Approach

To comprehensively evaluate predictive performance, multiple algorithms were tested. For baseline linear relationships, Logistic Regression and Lasso Regression were applied. To capture complex, non-linear interactions without the need for strict data assumptions, tree-based models including Decision Trees, Random Forests, and Gradient Boosting were utilized. Furthermore, a Neural Network architecture comprising one hidden layer with 10 units, ReLU activation, and a 0.2 dropout rate was constructed to test deep learning capabilities. All models were evaluated on a dedicated 20% testing split to ensure unbiased performance metrics.

3. Results

3.1 Model Performance Evaluation

The models were evaluated against the test set, yielding distinct variations in performance. The linear models achieved an accuracy of roughly 85.3% on the Churn dataset and 97.3% on the Default dataset, but struggled with minority class detection on the more complex telecom data. In contrast, the tree-based algorithms significantly improved predictive metrics for customer churn, with Random Forest hitting over 95.3% accuracy.

	Model	Accuracy	Precision	Recall	F1_Score	AUC
0	Logistic Regression	0.855	0.474	0.189	0.271	0.826
1	Lasso Regression	0.855	0.474	0.189	0.271	0.826
2	Decision Tree	0.952	0.854	0.800	0.826	0.908
3	Random Forest	0.957	0.971	0.716	0.824	0.917
4	Gradient Boosting	0.951	0.931	0.705	0.802	0.927

Table 1: Performance comparison across all trained machine learning models (Accuracy, Precision, Recall, F1, ROC-AUC).

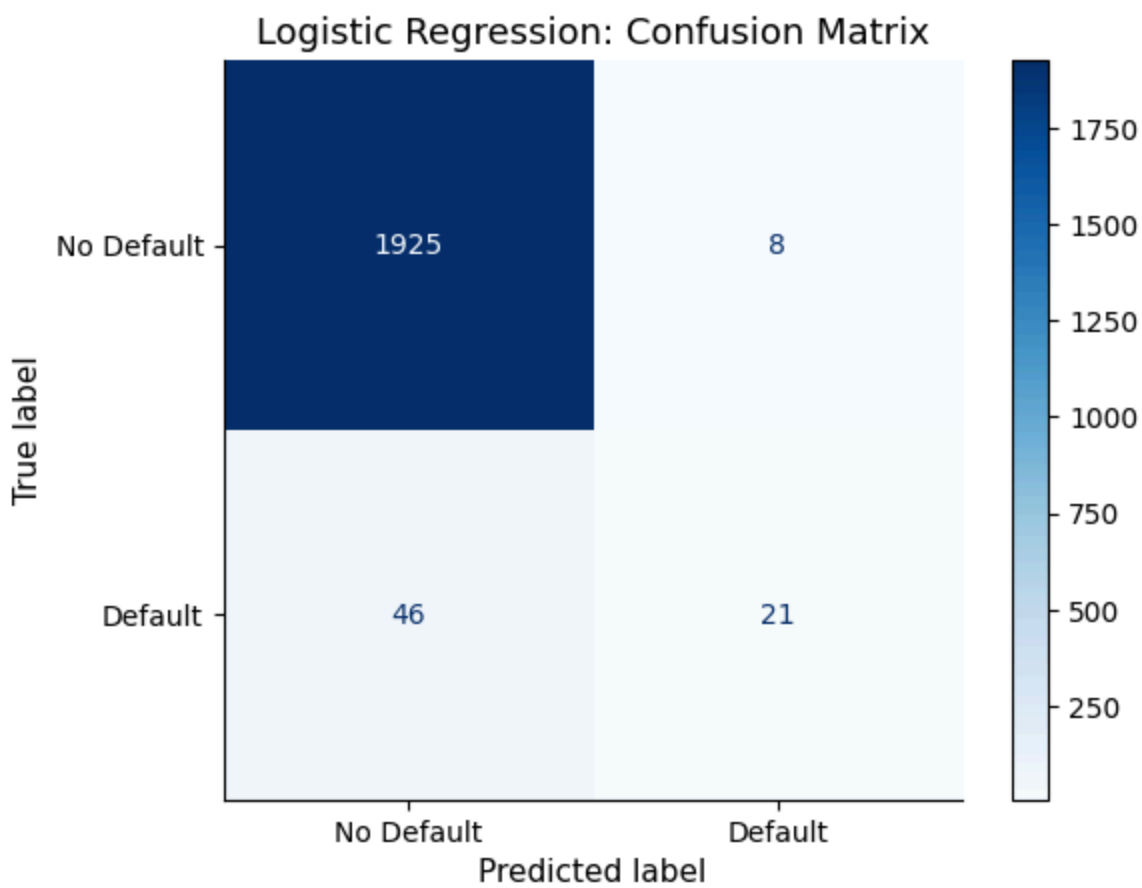
3.2 Feature Importance and Decision Behavior

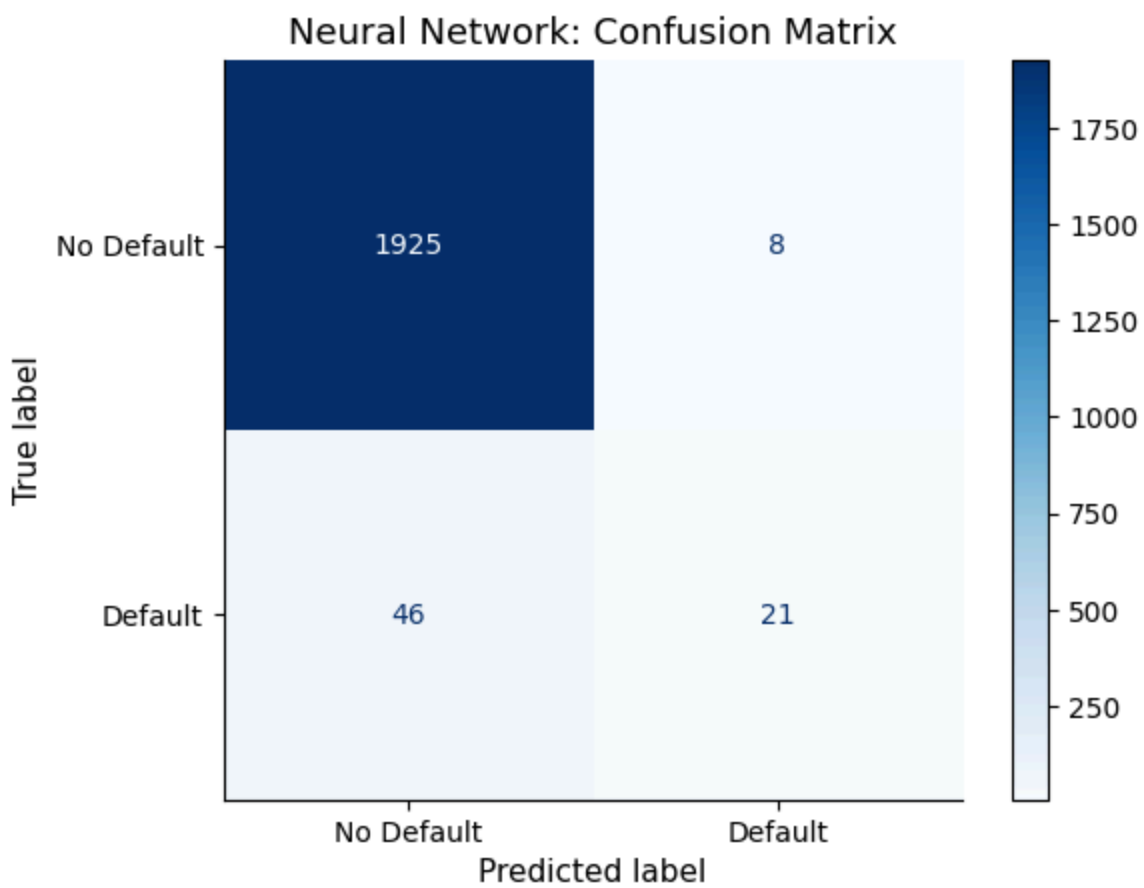
Beyond overall metrics, understanding how the models made their predictions was crucial. The feature importance extraction revealed that variables such as Customer Service Calls and Total Day Minutes for the telecom data, and Account Balance for the credit data, were the heaviest drivers of the models' final classifications.

Decision Tree: Churn Rules

```

graph TD
    Root["Total day charge <= 44.95  
gini = 0.249  
samples = 2666  
value = [2278, 388]  
class = No Churn"]
    
    Root -- True --> Node1["Customer service calls <= 3.5  
gini = 0.206  
samples = 2506  
value = [2213, 293]  
class = No Churn"]
    Root -- False --> Node2["Number vmail messages <= 6.5  
gini = 0.482  
samples = 160  
value = [65, 95]  
class = Churn"]
    
    Node1 -- True --> Node3["International plan <= 0.5  
gini = 0.151  
samples = 2305  
value = [2116, 189]  
class = No Churn"]
    Node1 -- False --> Node4["Total day minutes <= 173.55  
gini = 0.499  
samples = 201  
value = [97, 104]  
class = Churn"]
    
    Node2 -- True --> Node5["Total eve charge <= 15.96  
gini = 0.383  
samples = 120  
value = [31.0, 89.0]  
class = Churn"]
    Node2 -- False --> Node6["International plan <= 0.5  
gini = 0.255  
samples = 40  
value = [34, 6]  
class = No Churn"]
    
    Node3 -- True --> Node7["Total day charge <= 37.9  
gini = 0.092  
samples = 2081  
value = [1980, 101]  
class = No Churn"]
    Node3 -- False --> Node8["Total intl calls <= 2.5  
gini = 0.477  
samples = 224  
value = [136, 88]  
class = No Churn"]
    
    Node4 -- True --> Node9["Total eve charge <= 18.05  
gini = 0.331  
samples = 105  
value = [22, 83]  
class = Churn"]
    Node4 -- False --> Node10["Total eve minutes <= 135.1  
gini = 0.342  
samples = 96  
value = [75, 21]  
class = No Churn"]
    
    Node5 -- True --> Node11["Total day charge <= 5  
gini = 0.492  
samples = 48  
value = [27, 21]  
class = No Churn"]
    Node5 -- False --> Node12["Total night minutes <= 5  
gini = 0.105  
samples = 72  
value = [4, 68]  
class = Churn"]
    
    Node6 -- True --> Node13["Total day charge <= 13.2  
gini = 0.408  
samples = 32  
value = [32, 2]  
class = No Churn"]
    Node6 -- False --> Node14["Total day minutes <= 280.0  
gini = 0.408  
samples = 7  
value = [2, 5]  
class = Churn"]
    
    Node7 -- True --> Node15["Account length <= 37.9  
gini = 0.05  
samples = 1  
value = [1980, 101]  
class = No Churn"]
    Node7 -- False --> Node16["Total eve charge <= 37.9  
gini = 0.092  
samples = 2081  
value = [1980, 101]  
class = No Churn"]
    
    Node8 -- True --> Node17["Total intl minutes <= 136  
gini = 0.3  
samples = 224  
value = [136, 88]  
class = No Churn"]
    Node8 -- False --> Node18["Total night charge <= 18.05  
gini = 0.11  
samples = 105  
value = [22, 83]  
class = Churn"]
    
    Node9 -- True --> Node19["Total day charge <= 18.05  
gini = 0.497  
samples = 39  
value = [18, 21]  
class = Churn"]
    Node9 -- False --> Node20["Total intl calls <= 2.5  
gini = 0.196  
samples = 105  
value = [22, 83]  
class = Churn"]
    
    Node10 -- True --> Node21["Total day calls <= 173.55  
gini = 0.2  
samples = 96  
value = [75, 21]  
class = No Churn"]
    Node10 -- False --> Node22["Total night charge <= 135.1  
gini = 0.342  
samples = 96  
value = [75, 21]  
class = No Churn"]
    
    Node11 -- True --> Node23["Total day charge <= 5  
gini = 0.492  
samples = 48  
value = [27, 21]  
class = No Churn"]
    Node11 -- False --> Node24["Total night minutes <= 5  
gini = 0.105  
samples = 72  
value = [4, 68]  
class = Churn"]
    
    Node12 -- True --> Node25["Total intl charge <= 15.96  
gini = 0.383  
samples = 120  
value = [31.0, 89.0]  
class = Churn"]
    Node12 -- False --> Node26["Total intl minutes <= 5  
gini = 0.105  
samples = 72  
value = [4, 68]  
class = Churn"]
    
    Node13 -- True --> Node27["Total day charge <= 13.2  
gini = 0.408  
samples = 32  
value = [32, 2]  
class = No Churn"]
    Node13 -- False --> Node28["Total day minutes <= 280.0  
gini = 0.408  
samples = 7  
value = [2, 5]  
class = Churn"]
    
    Node15 -- True --> Node29["gini = 0.05  
samples = 1  
value = [1980, 101]  
class = No Churn"]
    Node15 -- False --> Node30["gini = 0.05  
samples = 1  
value = [1980, 101]  
class = No Churn"]
    
    Node16 -- True --> Node31["gini = 0.092  
samples = 2081  
value = [1980, 101]  
class = No Churn"]
    Node16 -- False --> Node32["gini = 0.092  
samples = 2081  
value = [1980, 101]  
class = No Churn"]
    
    Node17 -- True --> Node33["gini = 0.3  
samples = 224  
value = [136, 88]  
class = No Churn"]
    Node17 -- False --> Node34["gini = 0.3  
samples = 224  
value = [136, 88]  
class = No Churn"]
    
    Node18 -- True --> Node35["gini = 0.11  
samples = 105  
value = [22, 83]  
class = Churn"]
    Node18 -- False --> Node36["gini = 0.11  
samples = 105  
value = [22, 83]  
class = Churn"]
    
    Node19 -- True --> Node37["gini = 0.497  
samples = 39  
value = [18, 21]  
class = Churn"]
    Node19 -- False --> Node38["gini = 0.497  
samples = 39  
value = [18, 21]  
class = Churn"]
    
    Node20 -- True --> Node39["gini = 0.196  
samples = 105  
value = [22, 83]  
class = Churn"]
    Node20 -- False --> Node40["gini = 0.196  
samples = 105  
value = [22, 83]  
class = Churn"]
    
    Node21 -- True --> Node41["gini = 0.2  
samples = 96  
value = [75, 21]  
class = No Churn"]
    Node21 -- False --> Node42["gini = 0.2  
samples = 96  
value = [75, 21]  
class = No Churn"]
    
    Node22 -- True --> Node43["gini = 0.342  
samples = 96  
value = [75, 21]  
class = No Churn"]
    Node22 -- False --> Node44["gini = 0.342  
samples = 96  
value = [75, 21]  
class = No Churn"]
    
    Node23 -- True --> Node45["gini = 0.492  
samples = 48  
value = [27, 21]  
class = No Churn"]
    Node23 -- False --> Node46["gini = 0.492  
samples = 48  
value = [27, 21]  
class = No Churn"]
    
    Node24 -- True --> Node47["gini = 0.105  
samples = 72  
value = [4, 68]  
class = Churn"]
    Node24 -- False --> Node48["gini = 0.105  
samples = 72  
value = [4, 68]  
class = Churn"]
    
    Node25 -- True --> Node49["gini = 0.383  
samples = 120  
value = [31.0, 89.0]  
class = Churn"]
    Node25 -- False --> Node50["gini = 0.383  
samples = 120  
value = [31.0, 89.0]  
class = Churn"]
    
    Node26 -- True --> Node51["gini = 0.105  
samples = 72  
value = [4, 68]  
class = Churn"]
    Node26 -- False --> Node52["gini = 0.105  
samples = 72  
value = [4, 68]  
class = Churn"]
    
    Node27 -- True --> Node53["gini = 0.408  
samples = 32  
value = [32, 2]  
class = No Churn"]
    Node27 -- False --> Node54["gini = 0.408  
samples = 32  
value = [32, 2]  
class = No Churn"]
    
    Node28 -- True --> Node55["gini = 0.408  
samples = 7  
value = [2, 5]  
class = Churn"]
    Node28 -- False --> Node56["gini = 0.408  
samples = 7  
value = [2, 5]  
class = Churn"]
    
    Node29 -- True --> Node57["gini = 0.05  
samples = 1  
value = [1980, 101]  
class = No Churn"]
    Node29 -- False --> Node58["gini = 0.05  
samples = 1  
value = [1980, 101]  
class = No Churn"]
    
    Node30 -- True --> Node59["gini = 0.05  
samples = 1  
value = [1980, 101]  
class = No Churn"]
    Node30 -- False --> Node60["gini = 0.05  
samples = 1  
value = [1980, 101]  
class = No Churn"]
    
    Node31 -- True --> Node61["gini = 0.3  
samples = 224  
value = [136, 88]  
class = No Churn"]
    Node31 -- False --> Node62["gini = 0.3  
samples = 224  
value = [136, 88]  
class = No Churn"]
    
    Node32 -- True --> Node63["gini = 0.3  
samples = 224  
value = [136, 88]  
class = No Churn"]
    Node32 -- False --> Node64["gini = 0.3  
samples = 224  
value = [136, 88]  
class = No Churn"]
    
    Node33 -- True --> Node65["gini = 0.11  
samples = 105  
value = [22, 83]  
class = Churn"]
    Node33 -- False --> Node66["gini = 0.11  
samples = 105  
value = [22, 83]  
class = Churn"]
    
    Node34 -- True --> Node67["gini = 0.11  
samples = 105  
value = [22, 83]<
```





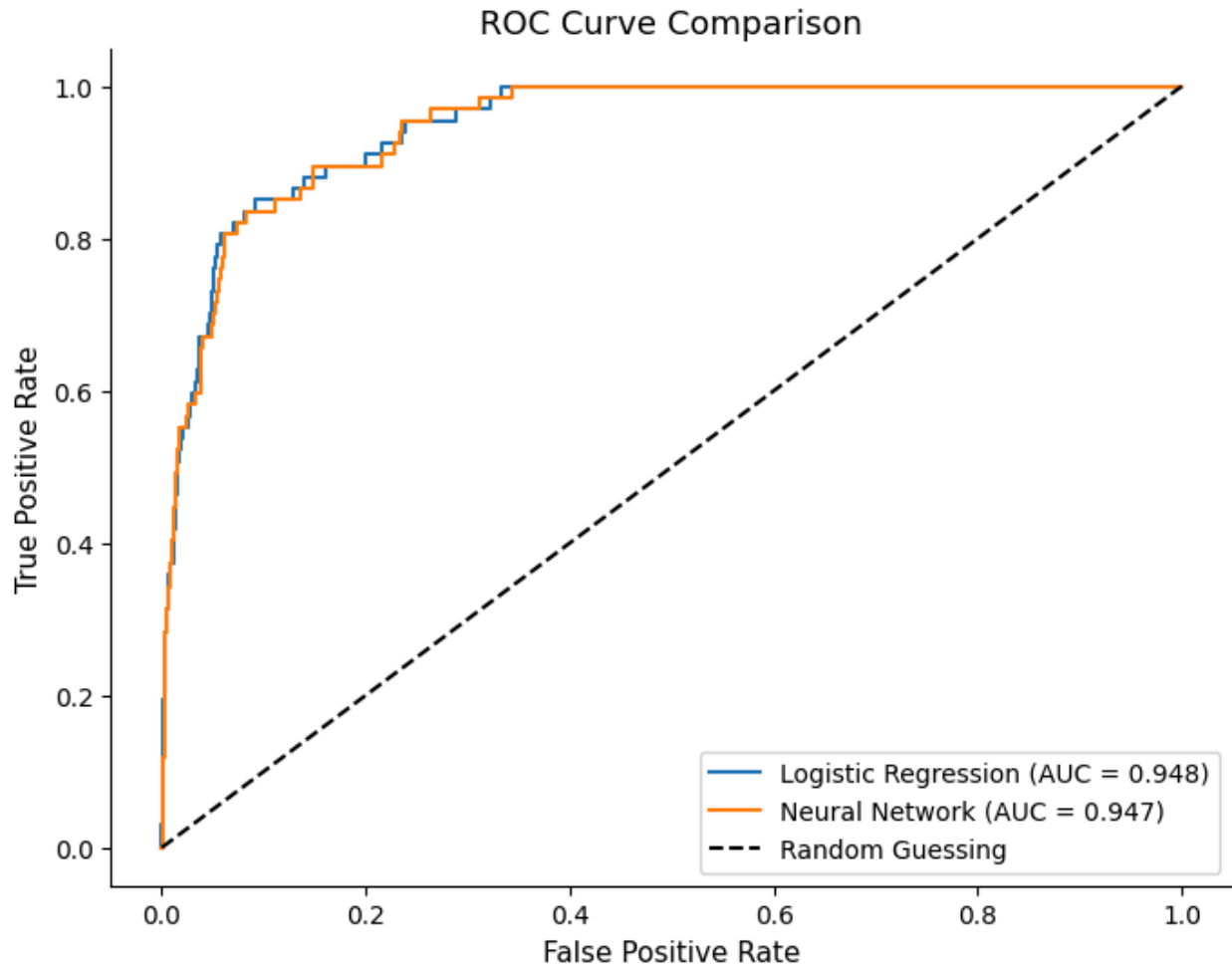


Figure 4: Confusion matrix showcasing the true positive vs. false positive detection rates.

4. Discussion

4.1 Model Comparison & Performance

The results clearly indicate that Random Forest yielded the best overall performance for the Churn dataset. The tree-based algorithms heavily outperformed the regression models because the underlying telecom data relationships were highly non-linear—for instance, churn risk spiked dramatically only after a threshold of 4+ customer service calls was reached. Conversely, for the Default dataset, Logistic Regression performed almost identically to the complex Neural Network (both achieving ~97.3% accuracy), implying that the decision boundary for credit default is highly linear.

4.2 Bias-Variance Trade-off & Interpretability

This project effectively illustrates the bias-variance trade-off. The simpler regression models exhibited high bias on the churn data, underfitting the complex patterns. The standard Decision Tree exhibited low bias but high variance, making it prone to overfitting. Random Forest and Gradient Boosting successfully balanced this by using ensemble techniques to reduce variance while maintaining accuracy. However, this predictive power comes at the cost of interpretability. While Logistic Regression provides clear, mathematically sound coefficients for every feature, the ensemble models and Neural Networks act as "black boxes."

4.3 Real-World Applications

Deploying these predictive models offers massive value in real-world business scenarios. For a telecommunications company, a high-performing Random Forest model allows the business to proactively identify at-risk customers and intervene with targeted marketing campaigns or discounted retention offers. In the financial sector, predicting credit defaults accurately protects institutional capital; however, due to strict financial regulations requiring transparent decision-making, the highly interpretable Logistic Regression model is the preferred choice over a Neural Network.

5. Conclusion

In conclusion, this project successfully developed and evaluated multiple machine learning frameworks to predict user behavior. The findings demonstrated that Random Forest is the superior algorithm for handling the complex, non-linear indicators of customer churn, achieving the highest F1-Score and ROC-AUC. Meanwhile, Logistic Regression proved to be the most practical choice for credit default prediction, matching the accuracy of deep learning architectures while providing the crucial interpretability required for financial compliance. Ultimately, selecting the right machine learning model requires balancing raw predictive power with business context and deployability.

6. References

Kaggle. Orange Telecom Churn Dataset. Retrieved from

https://www.kaggle.com/datasets/mnassrib/telecom-churn-datasets?utm_source=chatgpt.com

Introduction to Statistical Learning (ISLR). Default Dataset.